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Modular axle clamp and riser system

Abstract

A modular axle clamp and riser system for use in a suspension system, which includes a top mount having a substantially flat upper surface, a top mount body which extends transversely to the flat upper surface, a pair of opposing ears which extend from the top mount body, and at least one top mount tooth extending from the top mount body in a direction opposite the ears. Also included is a wedge with at least one wedge tooth which matingly engages the at least one top mount tooth, a wedge upper surface, and a pinion angle surface. Further, the wedge defines a pinion angle between the wedge upper surface and the pinion angle surface. Moreover, an axle mount with at least one axle mount tooth which matingly engages at least one of the at least one top mount tooth and the at least one wedge tooth is provided.

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Background/Summary

RELATED APPLICATION (1) This application claims priority under 35 USC 119 from U.S. Provisional application Ser. No. 63/261,246 filed Sep. 15, 2021, the entire contents of which are incorporated by reference herein.

BACKGROUND
(1) The present disclosure relates generally to vehicle suspensions used on heavy duty trucks, buses, recreational vehicles and the like, and more specifically to clamping systems for securing the springs to the suspension.
(2) An important part of any vehicle suspension system is the clamping system used in the suspension system. However, these clamping systems are particularly important for heavy vehicles, where the suspension system experiences significant loading. Additionally, there are many different configurations for suspension systems, as many auto manufacturers have specialized parts that are specific to the vehicles they manufacture. As such, it is difficult to supply clamping systems that can complement all of the various axle designs that exist. Even minor variations in the design of the axle or other components of the suspension system can necessitate a completely different clamping

system. In the past, manufacturers of such clamping systems have had to specially assemble and/or manufacture each of the multi-component clamping systems to meet performance and handling requirements in a wide variety of suspension systems produced by auto manufacturers. This often results in having to inventory or monitor and/or produce thousands of different clamping systems.

(3) In particular, each variation in the clamping system involves different production tooling. Whether the variation is adjusting the location of a through hole or adjusting the angle of one of the internal components, each minor change also involves unique and/or specialized machining. In turn, these minor changes have to be designed first, which takes significant effort. Then, the required tooling needs to be prepared. For low quantity orders, this process is not cost effective. However, with the wide range of variations that exist in suspension systems, there are many times where orders for a particular configuration of a clamping system may only be used by a handful of vehicles. Therefore, the manufacturer is faced with the decision of whether to produce a few clamping systems for which they may not be able to generate a profit or refuse to supply the parts and risk losing potential customers.

(4) Additionally, many operators of heavy vehicles prefer that the heavy vehicle suspension system be designed such that the ride height is increased, with the body of the vehicle farther from the ground. To accomplish this, the clamping system typically includes spacers. However, the inclusion of these spacers, with the fact that certain customers prefer different levels of ride height, provides another variable design parameter that is accounted for in producing clamping systems. Accordingly, there is the need for a clamping system which allows for accommodation of the many variations that exists in suspension systems without the significant cost and effort that is required at present.

SUMMARY

(5) The above listed need is met or exceeded by the present modular axle clamp and riser system. An important feature of the present disclosure is the use of interchangeable, easily assembled components which make it significantly more cost effective to accommodate the many variations in clamping systems that are requested by vehicle manufacturers. A source of expense associated with non-interchangeable clamping systems is caused by the need for minor variation in one particular part of the clamping system. For instance, a guide slot located at the bottom of the clamping system may need to be offset for a given customer. Under current systems, this would potentially necessitate generating an entirely clamping system. However, with the interchangeable components of the present disclosure, only the single component which houses the guide slot needs to be adjusted to satisfy the specific configuration requested by the consumer. In the same fashion, each of the components has a relatively small number of variations needed to cover a wide range of potential design configurations.

(6) As further illustration, if there were six exemplary total variations in each of the three standard components that make up the presently disclosed modular axle clamp and riser system, there would be a total of 216 combinations of the modular axle clamp and riser system that could be achieved. Previously, 216 different parts would be needed to cover each of these combinations, each part being separately designed, manufactured and potentially inventoried. With the presently disclosed innovative design, a total of eighteen individual components can accomplish the same number of clamping system variations.

(7) In addition to significantly reducing the cost and effort associated with manufacturing various modular axle clamp and riser systems, the present disclosure also provides the possibility of implementing a smart part numbering system, where each part is assigned a unique part number. In particular, each component of the modular axle clamp and riser system has a component level digit which specifies the particular component type, and at least one variation level digit which is associated with various design features of the components. Therefore, each set of design parameters for a given component is denoted with a single unique part number which includes both of the component level digit and the at least one variation level digit. This allows a consumer to

choose their unique combination of components to build the desired modular axle clamp and riser system, thereby significantly increasing the convenience for the consumer. Specifically, the consumer can either select the particular components based on the unique part numbers for each part, or the consumer can provide the desired design parameters, and the smart part numbering system will provide the requested modular axle clamp and riser system.

(8) Moreover, the interchangeable components of the present modular axle clamp and riser system are designed to easily fit together, thereby reducing the cost of assembling and welding the clamping system by providing the possibility of robotically welded the components. Additionally, the interface between the interchangeable components is designed to include either chamfered or filleted edges, thereby creating a groove where the parts are welded together. This allows the weld material to penetrate into the seem between the parts, and it greatly reduces the needed weld prep, as the groove provides the path for welding.

(9) Another benefit of the presently disclosed modular axle clamp and riser system is that the amount of post-fabrication machining is significantly reduced. Specifically, the individual components are designed such that they need not be machined, and upon assembly, they are welded together. As machining is a considerable expense, removing this step in the manufacturing process significantly reduces cost. Further, the present disclosure includes an axle mount that constitutes the bottom component of the modular axle clamp and riser system. A common requirement for the axle mount includes an offset guide slot. With the currently disclosed assembly, the axle mount is symmetrical from front to back, providing the ability to simply rotate the axle mount with respect to the rest of the clamping system, thereby reducing the number of variations needed for the axle mount. Additionally, a pinion angle of the modular axle clamp and riser system is adjustable by simply adjusting a single of the interchangeable components.

(10) More specifically, the present disclosure includes a modular axle clamp and riser system for use in a suspension system, which includes a top mount having a substantially flat upper surface, a top mount body which extends transversely to the flat upper surface, a pair of opposing ears which extend from the top mount body and include eyelets for receiving a torque arm of the suspension system, and at least one top mount tooth extending from the top mount body in a direction opposite the ears. Also included is a wedge with at least one wedge tooth which matingly engages the at least one top mount tooth, a wedge upper surface, and a pinion angle surface. Further, the wedge defines a pinion angle between the wedge upper surface and the pinion angle surface. Additionally, the wedge upper surface is parallel to the top mount flat upper surface. Moreover, the present modular axle clamp and riser system includes an axle mount with at least one axle mount tooth which matingly engages at least one of the at least one top mount tooth and the at least one wedge tooth.

(11) In an embodiment, the axle mount has a guide slot configured for receiving a dowel pin connected to an axle of said suspension system. An alternate embodiment includes a lower spacer located between the top mount and the wedge, such that the lower spacer includes at least one lower spacer tooth, where the lower spacer tooth is configured to matingly engage with at least one of the at least one top mount tooth and the at least one wedge tooth. In yet another embodiment, the modular axle clamp and riser system has an upper spacer which includes a locating post configured to fit within a locating hole in the top mount flat upper surface.

(12) Preferably, the teeth of the top mount, the wedge and the axle mount are welded together. Preferably still, the at least one axle mount tooth is directly in front of the at least one wedge tooth, the at least one axle mount tooth fits within at least one top mount recess, and the at least one top mount tooth fits within at least one axle mount recess, whereby the at least one wedge tooth is located between the at least one axle mount tooth and the at least one top mount recess. In an embodiment, the at least one axle mount tooth and the at least one axle mount cavity alternate with the at least top mount tooth and the at least one top mount cavity. Another preferred embodiment includes an interface where the teeth matingly engage one another, and the teeth have rounded

edges at each point along the interface, such that said rounded edges form a passageway for the weld material when the top mount, the wedge, and the axle mount are assembled. Preferably, the interface between the teeth of the top mount, the wedge and the axle mount includes at least one segment which is parallel to the axis which runs through the teeth.
(13) Preferably, the wedge and the axle mount have a second set of mating teeth, the torque pin is connected to the eyelets by one of a bar pin, a through bolt, or an hourglass bushing, and the dowel pin is welded to the axle.
(14) A second embodiment of the present disclosure is a smart part numbering system for a modular axle clamp and riser system, which includes a computer processor and a non-transitory, computer-readable medium encoded with computer-readable instructions. When these instructions are executed by the computer processor, the system generates a unique part number for each component of the modular axle clamp and riser system. In particular, the part number includes a component level digit which specifies the particular component type, and at least one variation level digit which is associated with various design features of the components, such that each set of design parameters for a given component are denoted with a single unique part number which includes the component level digit and the at least one variation level digit. Further, the instructions store the unique part numbers in a part library along with the specific design parameters associated with the unique part number. Additionally, the instructions receive one or more user inputs, each user input indicating a selection of either the design parameters or the unique part number combination desired for the modular axle clamp and riser system. Moreover, the instructions determine a requested modular axle clamp and riser system based on either the design parameters or the unique part number combination provided by the user and generate the requested modular axle clamp and riser system.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a side perspective view of a suspension system which includes the present modular axle clamp and riser system;
(2) FIG. **2** is a side perspective view of a prior art clamping system with spacers used in a suspension system;
(3) FIG. **3** is a side perspective view of the present modular axle clamp and riser system with upper and lower spacers;
(4) FIG. **4** is a side perspective view of a prior art clamping system without spacers;
(5) FIG. **5** is a side perspective view of the present modular axle clamp and riser system without spacers;
(6) FIG. **6** is a side exploded perspective view of the present modular axle clamp and riser system without spacers;
(7) FIG. **7** is a bottom perspective view of the present modular axle clamp and riser system without spacers;
(8) FIG. **8** is a bottom exploded perspective view of the present modular axle clamp and riser system without spacers;
(9) FIG. **9** is a side perspective view of the present modular axle clamp and riser system with the lower spacer;
(10) FIG. **10** is a side exploded perspective view of the present modular axle clamp and riser system with the lower spacer;
(11) FIG. **11** is a bottom perspective view of the present modular axle clamp and riser system with the lower spacer;
(12) FIG. **12** is a bottom exploded perspective view of the present modular axle clamp and riser

system with the lower spacer;
(13) FIG. **13** is a bottom perspective view of the present modular axle clamp and riser system without spacers;
(14) FIG. **14** is a top perspective view of two axle mounts of the present modular axle clamp and riser system;
(15) FIG. **15** is a bottom plan view of the two axle mounts of the modular axle clamp and riser system from FIG. **15**;
(16) FIG. **16** is a bottom plan view of two axle mount of the present modular axle clamp and riser system with an offset guide slot;
(17) FIG. **17** is a bottom plan view of the axle mount of the present modular axle clamp and riser system;
(18) FIG. **18** is a side plan view of wedges of the present modular axle clamp and riser system which have different pinion angles;
(19) FIG. **19** is side perspective view of the wedge the present modular axle clamp and riser system;
(20) FIG. **20** is side perspective view of the wedge the present modular axle clamp and riser system;
(21) FIG. **21** is side perspective view of a top mount the present modular axle clamp and riser system;
(22) FIG. **22** is side perspective view of the top mount the present modular axle clamp and riser system;
(23) FIG. **23** is side perspective view of the lower spacer the present modular axle clamp and riser system; and
(24) FIG. **24** is side perspective view of a manufacturing process of the present modular axle clamp and riser system; and
(25) FIG. **25** is a side exploded plan view a smart part numbering system for the present modular axle clamp and riser system.

DETAILED DESCRIPTION

(26) Referring now to FIGS. **1-23**, the present modular axle clamp and riser system is generally designated **10** and is shown in conjunction with a suspension system **12**. As shown in FIGS. **3** and **6**, the modular axle clamp and riser system **10** includes a top mount **14**, a wedge **16**, and an axle mount **18**. The top mount **14** includes a substantially flat upper surface **20**, a body **22**, and a pair of opposing ears **24**. Additionally, the ears **24** include eyelets **26** which accommodate a torque aim **28**. Preferably, the torque arm **28** is connected to the eyelets **26** by one of a bar pin, a through bolt, or an hourglass bushing. However, alternate means for connecting the torque arm **28** to the eyelets are contemplated as is known in the art.
(27) Moreover, the top mount **14** matingly engages the wedge **16** by way of top mount teeth **30** and wedge teeth **32**. Further, the wedge **16** includes a wedge upper surface **34** which is parallel to the top mount flat upper surface **20**. Importantly, the wedge includes a pinion angle surface **36**, such that a pinion angle α is defined as the angle between the pinion angle surface **36** and the top mount flat upper surface **20**. FIG. **18** illustrates three different pinion angles α, namely zero, four and six degrees. These particular pinion angles α are merely illustrative, as many different pinion angles α optionally are used.
(28) Additionally, the axle mount **18** has teeth **38** which matingly engage the at least one of the wedge teeth **32** and the top mount teeth **30**. An axle mount upper surface **40** has a guide slot **42** intended to accommodate a dowel pin (not shown) located on an axle **44**. The dowel pin is either welded or otherwise attached to the axle **44** as is known in the art. An important feature of the guide slot **42** is its location within the axle mount **18**. Specifically, the guide slot **42** is optionally offset from the center of the axle mount upper surface **40** depending on the application, as shown in FIG. **16**.

(29) In addition, the modular axle clamp and riser system optionally includes a lower spacer **46** which includes lower spacer teeth **48**. Preferably, the lower spacer teeth **48** matingly engage at least one of the top mount teeth **30** and the wedge teeth **32**. Another optional feature is an upper spacer **50**, which is best shown in FIG. **25**. Preferably, the upper spacer **50** includes a locating post **52** which fits within a locating hole **54** in the top mount **14**.

(30) A feature of the modular axle clamp and riser system **10** is an interface **56** which is formed by the mating engagement between the top mount teeth **30**, the wedge teeth **32**, and the axle mount teeth **38**. Moreover, rounded edges **58** of the top mount teeth **30**, the wedge teeth **32**, and the axle mount teeth **38** form a channel **60** at the interface **56**. This is significant for several reasons, the first of which is that it allows for a better weld, as the weld material is able to beyond the surface of the materials being welded. Moreover, the presence of the channel **60** means that the parts do not have to be pre-processed before they are welded. Traditionally, flat parts that are welded together need to be machined to create a groove into which the weld material will go. However, with preformed channels **60** created by the rounded edges **58** at the interface **56**, there is no need to prepare the parts for welding. Finally, with the channels **60** created by the rounded edges **58**, the welding can be done robotically.

(31) Along the interface **56**, there is at least one portion **62** which is transverse to a direction of loading for the modular axle clamp and riser system **10**. Including welding lines which are transverse to the direction of loading significantly increases the strength of the weld. In an example configuration, the at least one axle mount tooth **38** is directly in front of the at least one wedge tooth **32**, the at least one axle mount tooth **38** fits within at least one top mount recess **31**, and the at least one top mount tooth **30** fits within at least one axle mount recess **39**, whereby the at least one wedge tooth **32** is located between the at least one axle mount tooth **38** and the at least one top mount recess **31**. Further, the at least one axle mount tooth **38** and the at least one axle mount cavity **39** alternate with the at least top mount tooth **30** and the at least one top mount cavity **31**.

(32) FIG. **24** illustrates an exemplary means for manufacturing the modular axle clamp and riser system **10**. In particular, the top mount **14**, the wedge **16** and the axle mount **18** are assembled in the proper orientation and held against a table **70** by a pneumatic clamp **72**. Once the components of the modular axle clamp and riser system **10** are assembled and held in place by the table **70** and pneumatic clamp **72**, the components are welded together along the interface **56**. Other means for permanently affixing the components of the modular axle clamp and riser system **10** are also contemplated as is known in the art.

(33) FIG. **25** shows an example smart part numbering system **74** for the modular axle clamp and riser system **10**. Specifically, the smart part numbering system **74** includes a computer processor **76** which has a non-transitory, computer-readable medium encoded with computer-readable instructions that, are executed by the computer processor **76**. The instructions cause the computer processor **76** to generate a unique part number for each component of the modular axle clamp and riser system **10**. The unique part number includes a component level digit which specifies the particular component type, and at least one variation level digit which is associated with various design features of the components of the modular axle clamp and riser system **10**. Further, each set of design parameters for a given component are denoted with a single unique part number which includes the component level digit and the at least one variation level digit.

(34) Providing the present smart part numbering system **74** significantly increases the ease of manufacturing the modular axle clamp and riser system **10**. Welding of the modular axle clamp and riser system **10** can be done robotically, inventory management is simplified, and consumers have a less complicated process for ordering their requested the modular axle clamp and riser system **10**.

(35) While a particular embodiment of the present modular axle clamp and riser system has been described herein, it will be appreciated by those skilled in the art that changes and modifications may be made thereto without departing of the invention in its broader aspects and as set forth in the following claims.

Claims

1. A modular axle clamp and riser system for use in a suspension system, comprising: a top mount, comprising: a substantially flat upper surface; a top mount body which extends transversely to said flat upper surface; a pair of opposing ears which extend from said top mount body, said ears including eyelets for receiving a torque arm of said suspension system; and at least one top mount tooth extending from said top mount body in a direction opposite said ears; a wedge including at least one wedge tooth which matingly engages said at least one top mount tooth, a wedge upper surface, and a pinion angle surface, said wedge defining a pinion angle between said wedge upper surface and said pinion angle surface, said wedge upper surface being parallel to said top mount flat upper surface; and an axle mount comprising at least one axle mount tooth which matingly engages at least one of said at least one top mount tooth and said at least one wedge tooth.
2. The modular axle clamp and riser system of claim 1, wherein said axle mount further comprises a guide slot configured for receiving a dowel pin connected to an axle of said suspension system.
3. The modular axle clamp and riser system of claim 1, further comprising a lower spacer located between said top mount and said wedge, said lower spacer including at least one lower spacer tooth, said lower spacer tooth configured to matingly engage with at least one of said at least one top mount tooth and said at least one wedge tooth.
4. The modular axle clamp and riser system of claim 1, further comprising an upper spacer including a locating post which is configured to fit within a locating hole in said top mount flat upper surface.
5. The modular axle clamp and riser system of claim 1, wherein said teeth of said top mount, said wedge and said axle mount are welded together.
6. The modular axle clamp and riser system of claim 5, wherein said at least one axle mount tooth is directly in front of said at least one wedge tooth, said at least one axle mount tooth fits within at least one top mount recess, and said at least one top mount tooth fits within at least one axle mount recess, whereby said at least one wedge tooth is located between said at least one axle mount tooth and said at least one top mount recess, wherein said at least one axle mount tooth and said at least one axle mount cavity alternate with said at least top mount tooth and said at least one top mount cavity.
7. The modular axle clamp and riser system according to claim 6, such that said at least one axle mount tooth, said at least one wedge tooth, and said at least one top mount tooth are arranged along an axis which runs through said teeth.
8. The modular axle clamp and riser system of claim 1, further comprising an interface where said teeth matingly engage one another, wherein said teeth have rounded edges at each point along said interface, such that said rounded edges form a passageway for the weld material when said top mount, said wedge, and said axle mount are assembled.
9. The modular axle clamp and riser system of claim 8, wherein said interface between said teeth of said top mount, said wedge and said axle mount includes at least one segment which is parallel to said axis which runs through said teeth.
10. The modular axle clamp and riser system of claim 1, wherein said wedge and said axle mount further comprise a second set of mating teeth.
11. The modular axle clamp and riser system of claim 1, wherein said torque arm is connected to said eyelets by one of a bar pin, a through bolt, or an hourglass bushing.
12. The modular axle clamp and riser system of claim 2, wherein said dowel pin is welded to said axle.